

RAHUL MITRA

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EDUCATION

National University of Singapore

Singapore

Bachelor of Engineering, Industrial Systems Engineering

Aug 2023 – Jul 2027

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277) — top student in a class of 24, NUS School of Computing.
- Relevant coursework: Operations Research, Simulation, Statistics for Engineering Applications, Industrial Systems Engineering, Artificial Intelligence, 3D Computer Vision.

EXPERIENCE

STMicroelectronics

Singapore

Operations Research (NUS System Design Project), Contract

Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

Amazon

Singapore

Robotics Vision Engineer (Blended AI+X)

Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.

Abbott Laboratories

Singapore

Operational AI Systems & Data Engineer (Contract)

Jun 2026 – Present

- Reverse-engineer four plants' APC steady-state optimizers from their Excel master files into a single HiGHS linear program with a tracking QP and PLC predictor, returning setpoints, binding constraints, and shadow prices at millisecond latency without a commercial solver license.

Abbott Laboratories

Singapore

Data Analytics and Project/Product Management Intern

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT optimizer for hybrid flow-shop scheduling; compared heuristic, MIP, and genetic-algorithm approaches and researched robust optimization.
- Engineered a non-destructive, 15-stage changeover-data pipeline to normalize operator- and machine-generated anomalies, validating it against five years of unseen raw data with zero execution failures.

People's Association

Singapore

Software Engineer (Citizen Developer)

Sep 2025 – Present

- Built an end-to-end digital garden-plot balloting platform spanning admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.

Republic of Singapore Navy

Singapore

Base Support Assistant

Jan 2022 – Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.

PROJECTS

Hybrid Flow-Shop Digital Twin Optimizer

2026

- Modeled hybrid flow-shop scheduling as a SimPy digital twin with explicit objectives and constraints, using CP-SAT, simulation, and deployment-oriented Python tooling.

Volt Pulse SG — SMU Hack For Cities 2026 (Top 8 of 50+ teams)

Jan 2026

- Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

IE2110 Operations Research Graph Optimization

2024

- Built LP, network-flow, and nonlinear programming models, including a Jupyter SSSP/APSP (Dijkstra) graph optimization for engineering decision support.

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre

Singapore

Elderly Care Volunteer

Sep 2025 – Present

- Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

SKILLS AND CERTIFICATIONS

Technical: CP-SAT, HiGHS linear and quadratic programming, SciPy, hybrid flow-shop scheduling, SimPy, digital-twin simulation, robust optimization, graph optimization, network flow, Dijkstra/SSSP/APSP, decision analytics, statistics, Python, Excel VBA

Certifications: Excel 2019 Advanced (Intellezy); NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)